MILITARY SPECIFICATION

HEATERS, CONVECTION, STEAM AND HOT WATER

This amendment forms part of Military Specification
MIL-H-3117C (SHIPS), 1 February 1962.

Page 1, paragraph 2.1: Under Military Specifications delete reference to MIL-Q-9858 and add "MIL-S-901-Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements For."

Page 1, paragraph 2.1: Add the following:

"PUBLICATIONS

"BUREAU OF SHIPS
"NAVSHIPS 0900-001-7000-Fabrication and Inspection of Brazed Piping Systems."

Page 2, paragraph 3.3: Add "The heater shall be designed such that it is capable of passing the high impact (H.I.) shock tests specified in MIL-S-901 for grade A, class I equipment."

Page 3, paragraph 3.3.7, lines 2 and 5: Delete "3/8-inch" and substitute "1/4-inch".

Page 3, paragraph 3.4: Delete and substitute:

"3.4 Welding and brazing. - The surfaces of all parts to be welded or brazed shall be free from rust, scale, paint, grease and other foreign matter. All welding shall be at least equal to that required by MIL-STD-278. Brazing procedures and qualification of welder (brazer) shall be in accordance with NAVSHIPS 0900-001-7000, as applicable."

Page 4, paragraph 4.1.1: Delete.

Page 4, paragraph 4.2: Add "In addition to these tests the size 20 heater shall be tested as specified in 4.2.3."

Page 4: Add paragraphs 4.2.3 and 4.2.3.1:

"4.2.3 Shock tests. - The convection heater shall be shock tested on the light weight machine and the tests shall be as prescribed for type A of MIL-S-901. A hydrostatic pressure of at least 35 psig shall be maintained on the heating element of the heater during the shock tests. The tests shall be conducted after the tests specified in 4.2.2. Evidence of fragmentation or missile effect of parts, or heating element leakage shall be cause for rejection.

"4.2.3.1 Upon completion of shock tests, the sample heater shall be returned to the supplier for his examination and disposition."

Page 6: Add paragraphs 6.4, 6.4.1 and 6.4.2:

"6.4 Design guides. -

"6.4.1 Shock. - The design information given in MIL-S-901 should be carefully considered in the design of this equipment. This information is also used by the drawing review activity as a factor in the approval of drawings prior to manufacture.

"6.4.2 Vibration. - All machinery and equipment installed aboard Naval ships will ordinarily be subject to varying frequencies and amplitudes of vibration, possibly for long periods of time, during which the machinery and equipment must continue to perform its normal function. Environmental vibration frequencies

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up to and including 33 cycles per second may be encountered. In designing this equipment the information given in MIL-STD-167 for environmental vibration and also internally excited vibration should be considered."

Page 6: Delete "Notice. -" in its entirety.

Preparing activity:
Navy - SH
(Project 4520-N011SH)

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